

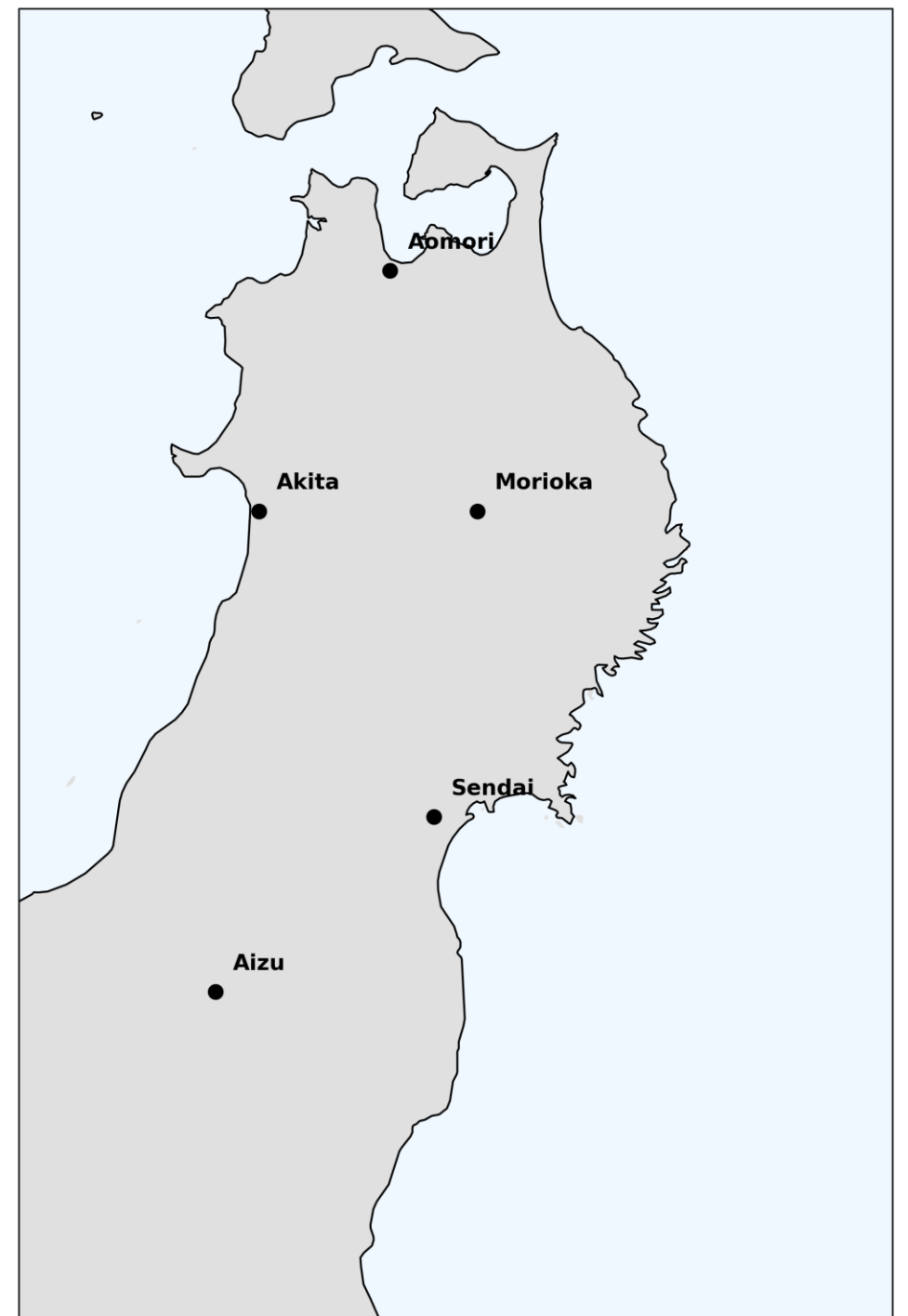
April 2026 Lab Seminar

Cloud-Aware HAP-Assisted Intercity QKD Sharing for Tohoku Area Networks

Koki Matano

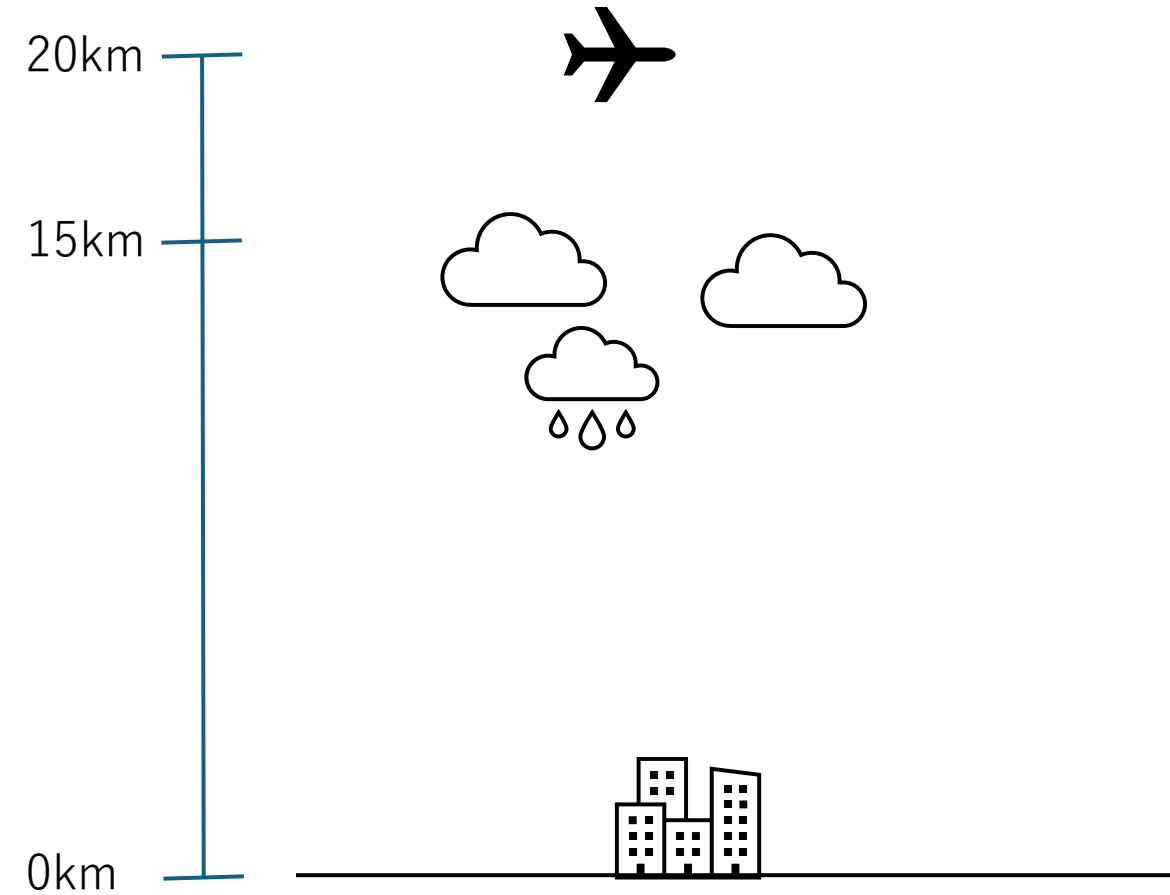
Research Motivation

- Critical Services require highly secure intercity communication
- Quantum Key Distribution (QKD) provides information-theoretic security
- **Challenge:** FSO links cannot penetrate thick clouds
- Tohoku has dense, seasonal cloud coverage
→ single-link QKD is unreliable



Why HAP?

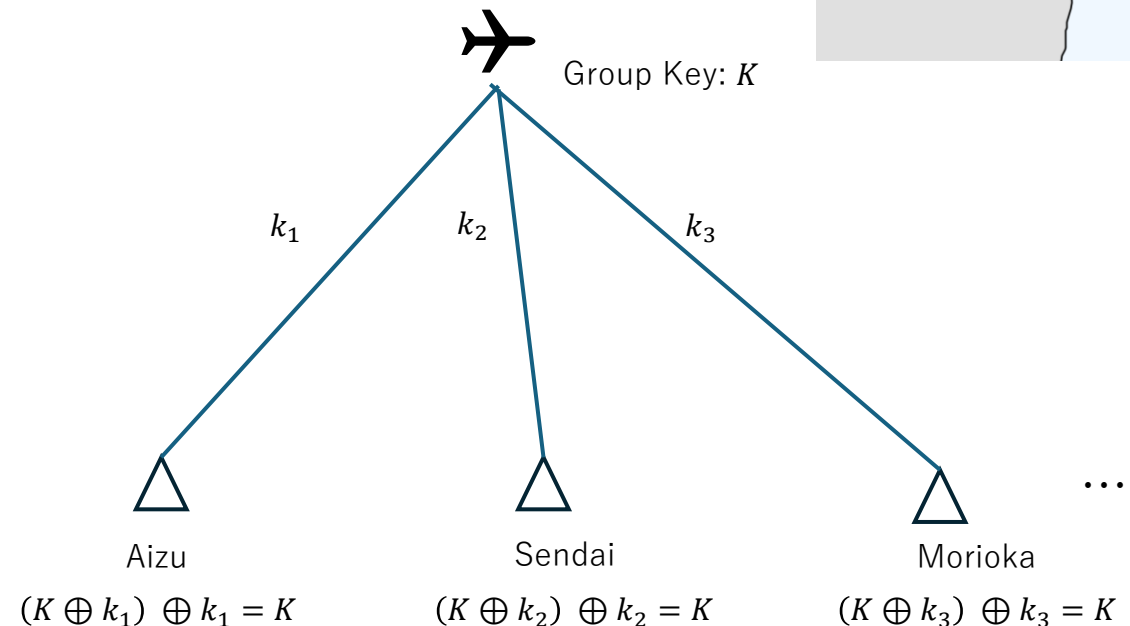
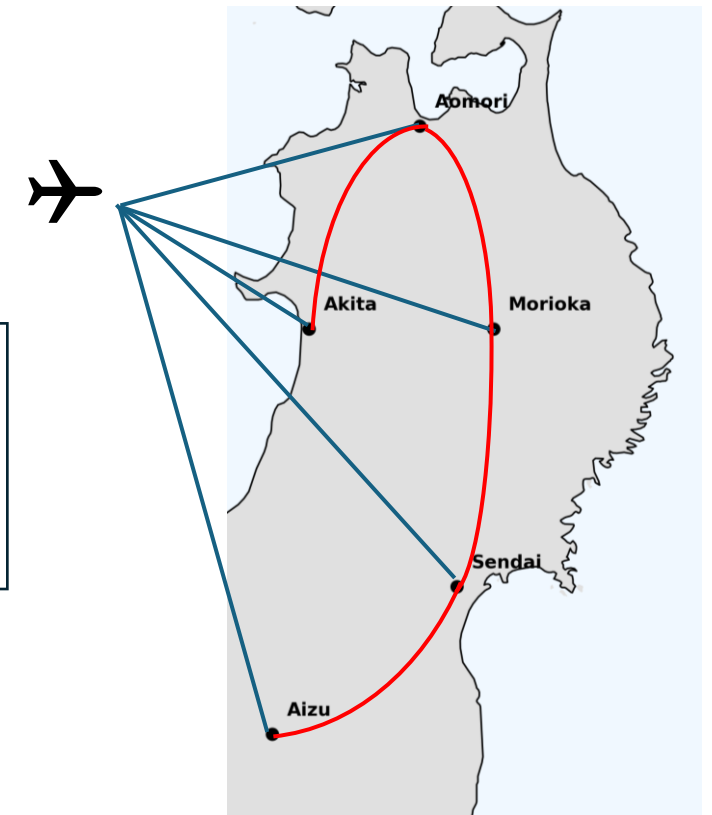
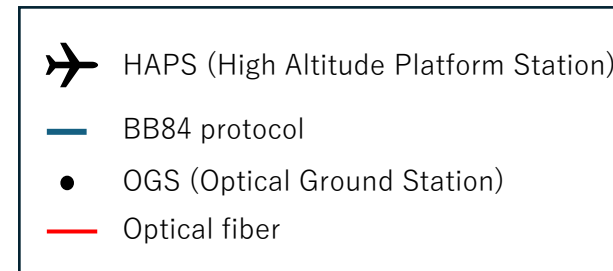
- Operates at 20 km altitude – above most clouds
- Lower path loss and latency than satellites
- Flexible deployment compared to fiber
- Wide-area coverage suitable for Tohoku's geography



System Model

- Single HAP, BB84 over FSO, time-division per city
- HAP generates pairwise keys k_i with each city
- HAP creates group key K , sends $K \oplus k_i$ to each city
- Cities recover K independently

Assumption: HAP is trusted / active cities only (SKR threshold)



Performance Metrics

- Link Availability — % of time each link is usable
- Average Active Cities — how many cities connected simultaneously
- Group Key Success Ratio — fraction of time $\geq k$ cities active

Evaluation Method

- Cloud data, 2024, hourly, 0.25° grid
- Per hour: compute attenuation → determine active cities → calculate 3 metrics
- 5 cities: Aomori, Akita, Morioka, Sendai, Aizuwakamatsu

Next Steps

Done

- System model and goals defined
- ERA5 cloud analysis implemented / SKR simulation under revision

Next

- Group Key Success Ratio calculation
- Full evaluation over 2024 dataset